MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY:1054S492
(OUTBOUND CONfigurATION)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-S0785 comprise the complete requirements for the Outbound Configuration Circuit Board Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation forbids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-S078S	Circuit Card Assemblies, General Specification for

DRAWMGS

U.S. Army, Frankford Arsenal

1054S425	Sight Current Generator
10545492	Circuit Card Assembly, Outbound Configuration

PACKAGING DATA SHEET

10545492	Packaging of Circuit Card Assembly, Outbound Configuration.

(Copies of specifications, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Item definition. This assembly is a single ended printed circuit board consisting of two relay driver circuits and seven relays connected in parallel. 'The assembly is used to control circuit routing when installed in the Sight current Generator, Drawing Number 10545425.

3.1.1 Major component list. The components of this assembly shall be as specified on the Engineering Parts List (EPL) 10545492.

3.1.2 Government furnished property list. Not applicable.

3.2 Characteristics

3.2.1 Performance. The assembly shall, when assembled in accordance with Drawing Number 10545492, be capable of perfoming the following functions:

a. Control relays K1 through K7 by means of the Q1 selay driver.

b. Control K21 relay mounted in the sight current generator through Q2 and Q3 relay drivers.

c. Reroute and/or ground as required, appropriate electrical circuits in the sight current generator, through K1 through K7 relays.

3.2.1.1 Input and Output levels

3.2.1.1.1 Input power. Input power shall be supplied to the assembly as specified in Table I.

Table I. Input Power

Pin Number	Voltage
4	+15 vdc ± 2.0 percent
14	-15 vdc ± 2.0 percent
24	+24 vdc ± 2.0 percent
1 and 28	Ground

3.2.1.1.2 The requirements specified in 3.2.1.1.2.1 through 3.2.1.: shall be met when supplied with the input voltages specified in Table I..

3.2.1.1.2.1 Pin 5 output. The output shall be +15 Vdc ± 10 percent between pin 5 and pin one with an input voltage of zero Vdc applied to pin 19. The output shall be zero ±0.5 Vdc between pin 5 and pin one with an input voltage of +15 Vdc applied to pin 19.

3.2.1.1.2.2 Pin 20 output. The output shall be zero ±.01 Vdc between pin 20 and pin 28 with an input voltage of +15 Vdc applied to pin 26. The output shall be +24 Vdc ±10 percent between pin 20 and pin 28 with an input voltage of zero Vdc applied to pin 26. The output shall be +24 Vdc ±10 percent between pin 20 and pin 28 with an input of +15 Vdc applied to pin 26 and +24 Vdc applied to pin 21. The output shall be +24 Vdc ± 10 percent between pin 20 and pin 28 with an input voltage of +15 Vdc applied to pin 26 and +24 Vdc applied to pin 23.

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3.2.1.1.2.3 Pin 21 output. The output shall be zero ± 0.5 Vdc between pin 21 and pin 28 with an input voltage of +15 Vdc applied to pin 26 and +24 Vdc applied to pin 23.

3.2.1.1.2.4 Pin 23 output. The output shall be zero ± 0.5 Vdc between pin 23 and pin 28 with an input voltage of +15 Vdc applied to pin 26 and +24 Vdc applied to pin 21.

3.2.1.2 Deenergized relay contact position. The denergized relay contact positions shall be as specified in Table II with no voltage applied to pin 19.

Table II. Relay Contact Positions

From Pin	To Pin	Contact Position
1	6, 7, 9, or 12	Open
2	8	Open
3	10	Open
11	13	Open

3.2.1.3 Energized relay contact position. The energized relay contact positions shall be as specified in Table III with +15 Vdc applied to pin 19.

Table III. Relay Contact Positions

From Pin	To Pin	Contact Position
1	6, 7, 9 or 12	Closed
2	8	Closed
3	10	Closed
11	13	Closed

3.2.1.4 Environmental. The assembly shall meet the environmental requirements of specification MIL-C-50785.

3.2.2 Physical characteristic. The physical Characteristics shall be as specified in drawing 10545492 and the engineering parts list.

3.3 Design and construction

3.3.1" Production drawing This assembly shall be fabricated in accordance with the requirements of specification MIL-C-50785, drawing 10545492 and associated lists.

3.3.2 Standards of manufacture. The standards of manufacture shall be those specified in specification MIL-C-59785.

3.3.3 Workmanship. The workmanship shall be as specified in specification MIL-C-50785.

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4. QUALITY ASSURANCE PROVISION

4.1 General

4.1.1 Responsibility for inspection. The responsibility for inspection shall be as specified in specification MIL-C-50785.

4.1.2 Special tests and examination. The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-50785.

4.2 Quality conformance inspections. The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-50785.

4.2.1 Test equipment

4.2.1.1 Power supply (PS_1). The power supply shall be 15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.2 Power supply (PS_2). The power supply shall be -15 ± 0.1 Vdc, 100 ma or greater, ripple less than 10 mv rms or 100 mv peak-to-peak.

4.2.1.3 Power supply (PS_3). The power supply shall be 24 ± 0.5 Vdc, 100 ma or greater, ripple less than 10 mv rms or 100 mv peak-to-peak.

4.2.1.4 Digital voltmeter. The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 Vdc. Input impedance greater than 10 megohms on all voltage scales.

4.2.1.5 Digital ohmmeter The digital ohmmeter shall have a resistance readout to three digits from zero ohms to 10,000,000 ohms with a 1.0 percent accuracy.

4.2.1.6 Resistor. One resistor is required with a rating of 1000 ohms ± 5.0 percent, 2.0 watt.

4.2.2 Functional tests

4.2.2.1 Test connections. The power supplies shall be connected to the assembly under test in accordance with Figure 1.

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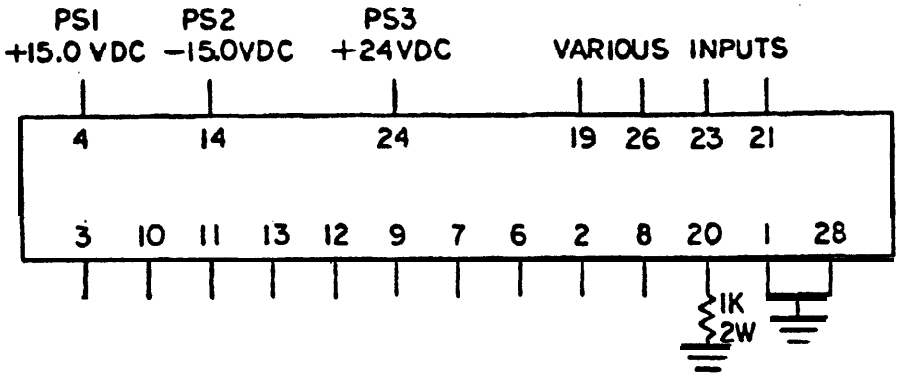


Figure 1. **Test Connections**

4.2.2.2 Test procedure. Compliance with the requirements of 3.2.1.1.2.1 through 3.2.1.5 shall be verified as follows:

a. With the inputs applied as specified in 3.2.1.1.2.1 and 3.2.1.1.2.4 the voltages measured with the DWM shall be as specified in those paragraphs.

b. with no voltage applied on pin 19, the resistance measured with the digital ohmmeter between each set of pins specified in Table 11 shall be 100,000 ohms minimum.

c. with +15 Vdc applied to pin 19, the resistance measured with the digital ohmeter between each set of pins specified in Table III shall be one ohm maximum.

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S. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging nd marking shall be in ccordance with the level A requirements of Packaging Data Sheet (PDS), AMC form 1029, 10545492,

6. NOTES:

6.1 The notes specified in the genera1 specification MIL-C-50785 are pplicable to this specification.

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Army-M'IJ

Preparing Activity

Army-MU
Project No. 1285-A3IO

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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